# Connector-Induced Representation Distortion and Semantic Stability in Vision–Language Models

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Code: https://github.com/drink36/VLM_Connectors

## Abstract

Projection connectors in vision–language models (VLMs) have been shown to lead to representation-level information loss and geometric distortion in visual representations (Li et al., 2025). Prior work suggests that these changes can be associated with downstream performance degradation, but this relationship appears to vary across model architectures.

In this work, we investigate whether connector-induced representation distortion is sufficient to explain semantic behavior in generated captions. We analyze pre- and post-projection representations through geometry-based comparison, reconstruction-based recoverability analysis, and controlled perturbation experiments.

Our results show that projection substantially alters the structure of the representation space. However, reconstruction-based measures of information loss do not consistently correlate with semantic changes in generated captions. Controlled perturbation experiments further show that caption outputs are sensitive to representation changes, but this sensitivity varies across models.

Overall, our findings suggest that representation distortion alone does not fully explain semantic behavior in VLMs. The relationship depends on model characteristics and on how task-relevant semantic information is organized in the projected representation space.

## 1 Introduction

Vision–language models (VLMs) often rely on a connector module to map visual representations into the embedding space of a language model. This projection step enables modality alignment, but it also changes the structure of the visual representation that the language model receives. As a result, the connector is not only a technical bridge between modalities, but also a transformation that may affect downstream language generation.

Prior work has shown that connector projection can alter representation geometry and lead to representation-level information loss. Li et al. (Li et al., 2025) analyze this problem using KNN overlap and embedding reconstruction. Their results show that geometry distortion is associated with retrieval degradation, and that reconstruction loss is negatively correlated with captioning and VQA performance for some models. However, this relationship is not consistent across all architectures: for example, Qwen-style models show large pre/post-projection changes but do not exhibit the same correlation between reconstruction loss and captioning behavior. This suggests that representation distortion alone may not be sufficient to explain downstream semantic behavior.

This motivates the question studied in this work: when are representation-level distortions reflected in caption semantics? Instead of assuming that higher reconstruction loss always implies semantic degradation, we revisit caption-level behavior using semantic-oriented metrics such as BERTScore (Zhang* et al., 2020) and CLIPScore (Hessel et al., 2021), together with controlled perturbation experiments. This analysis tests whether distortion-based measures provide a consistent account of caption-level semantic behavior across models.

To study this question, we analyze connector-induced changes at both the representation and caption levels. We first compare local neighborhood structure before and after projection, using CLIP-based neighborhoods as an external semantic reference. We then examine whether reconstruction-based information loss correlates with caption-level semantic metrics such as BERTScore and CLIPScore. Finally, we use controlled perturbation experiments to test whether changes in visual representations are reflected in generated captions. Together, these analyses examine how reliably representation distortion reflects caption-level semantic behavior.

## 2 Related Work

### 2.1 Connector-based VLM architectures

Modern VLMs differ substantially in how they transform visual features before passing them to the language model. LLaVA (Liu et al., 2023) uses a relatively direct MLP projector, IDEFICS2 (Laurençon et al., 2024) uses a Perceiver-style resampler, and Qwen2.5-VL (Bai et al., 2025) uses a patch-merging module. We also include Qwen3.5 (Team, 2026), a recent Qwen-family omnimodal model, as an exploratory case for testing whether the distortion–caption relationship extends to newer omnimodal architectures. These architectural differences matter because representation distortion may have different implications depending on whether the visual encoder and connector are frozen, jointly trained, or embedded within a broader omnimodal system.

### 2.2 Distortion and information loss

The closest work to ours is Li et al. (Li et al., 2025), which studies information loss in VLM connectors from the representation perspective. They quantify projection-induced changes using KNN overlap and embedding reconstruction, showing that connectors can substantially alter local representation geometry and reduce recoverability of pre-projection embeddings. They further connect these losses to downstream retrieval, captioning, and visually grounded question-answering behavior. However, their results also suggest that the relationship between representation distortion and downstream degradation is model-dependent, motivating further analysis of when such distortion is reflected in generated language.

### 2.3 Caption-level semantic evaluation

Captioning behavior is commonly evaluated using reference-based metrics such as CIDEr (Vedantam et al., 2015), which measures agreement between generated captions and human-written references. However, reference-based scores may not fully capture semantic stability between captions generated under different representation conditions. To complement this view, we use BERTScore (Zhang* et al., 2020) to measure text-level semantic similarity and CLIPScore (Hessel et al., 2021) to measure image-text alignment. These metrics allow us to examine whether representation-level distortion is reflected in caption semantics beyond a single reference-based quality score.

## 3 Analysis Framework

We analyze connector-induced representation changes from three perspectives: local geometry preservation, reconstruction-based information recoverability, and caption-level semantic behavior. We first define the pre- and post-projection representations, then describe the metrics used for each analysis.

### 3.1 Pre- and Post-Projection Representations

We use image captioning as a behavioral probe for analyzing whether representation-level changes are reflected in downstream language outputs. Given an input image $x$, a VLM first encodes the image into a sequence of visual representations:

$$z_{\text{pre}} = v(x),$$

where $v(\cdot)$ is the vision encoder. The connector then maps this representation into the language model embedding space:

$$z_{\text{post}} = p(z_{\text{pre}}),$$

where $p(\cdot)$ is the projection connector. The language model generates a caption conditioned on the projected representation:

$$y = D(z_{\text{post}}).$$

Our objective is not to improve caption quality, but to use caption generation as a functional readout of the projected visual representation. We therefore ask whether changes between $z_{\text{pre}}$ and $z_{\text{post}}$ are reflected in the semantics of the generated caption $y$. Throughout the analysis, $z_{\text{pre}}$ is treated as a reference representation, not as an absolute ground truth for visual semantics.

### 3.2 kNN-Based Geometry Preservation

We first analyze whether connector projection preserves the local geometry of visual representations. If the connector preserves local structure, images that are close in the pre-projection space should remain close after projection. Following Li et al. (Li et al., 2025), we measure this using the $k$-nearest-neighbor overlap ratio.

For each image $x_i$, we use the pre- and post-projection representations defined in Section 3.1:

$$z^i_{\text{pre}} = v(x_i), \qquad z^i_{\text{post}} = p(z^i_{\text{pre}}).$$

Since these representations are sequences of visual tokens, we obtain image-level vectors by mean pooling:

$$\bar{z}^i_{\text{pre}} = \text{pool}(z^i_{\text{pre}}), \qquad \bar{z}^i_{\text{post}} = \text{pool}(z^i_{\text{post}}).$$

We then compute the $k$ nearest neighbors of each image in the pre- and post-projection spaces, denoted as $\mathcal{N}_{\text{pre}}(i,k)$ and $\mathcal{N}_{\text{post}}(i,k)$. The pre/post neighborhood overlap is

$$R_{\text{pre,post}}(i,k) = \frac{|\mathcal{N}_{\text{pre}}(i,k) \cap \mathcal{N}_{\text{post}}(i,k)|}{k}.$$

A lower overlap indicates stronger local neighborhood reordering after projection, while a higher overlap suggests stronger preservation of local representation geometry.

However, low pre/post overlap does not necessarily imply semantic degradation. It only shows that the post-projection space differs from the original vision-encoder space. To better interpret this difference, we also compare both spaces against CLIP image embeddings as an external semantic reference. For each image, we compute a CLIP-based neighborhood $\mathcal{N}_{\text{clip}}(i,k)$ and measure

$$R_{a,\text{clip}}(i,k) = \frac{|\mathcal{N}_a(i,k) \cap \mathcal{N}_{\text{clip}}(i,k)|}{k}, \quad a \in \{\text{pre}, \text{post}\}.$$

If the post-projection neighborhood has higher overlap with CLIP than the pre-projection neighborhood, this suggests that projection may reorganize the representation space toward CLIP-style semantic similarity rather than simply degrading it. We use this comparison as a diagnostic reference rather than as ground-truth semantics.

### 3.3 Embedding Reconstruction

The kNN-based analysis measures whether connector projection preserves local neighborhood structure, but it does not directly measure how much pre-projection information remains recoverable after projection. To estimate this recoverability, we follow the reconstruction-based analysis of Li et al. (Li et al., 2025) and train a model to reconstruct pre-projection visual representations from post-projection representations.

Given the pre- and post-projection representations defined in Section 3.1,

$$z^i_{\text{pre}} = v(x_i), \qquad z^i_{\text{post}} = p(z^i_{\text{pre}}),$$

we train a reconstruction model $r_\theta$ to predict the original pre-projection representation:

$$\hat{z}^i_{\text{pre}} = r_\theta(z^i_{\text{post}}).$$

Following Li et al. (Li et al., 2025), we use either a 3-layer MLP reconstructor or an 8-layer Transformer reconstructor according to the reconstruction setting used for each model.

When the connector changes the number of visual tokens, the reconstruction model maps between different sequence lengths so that its output matches the shape of $z^i_{\text{pre}}$.

We define the reconstruction loss for image $x_i$ as

$$\mathcal{L}_{\text{rec}}(x_i) = \frac{1}{M_{\text{pre}}} \sum_{j=1}^{M_{\text{pre}}} \left\| z^i_{\text{pre},j} - \hat{z}^i_{\text{pre},j} \right\|_2^2,$$

where $j$ indexes visual tokens and $M_{\text{pre}}$ is the number of pre-projection visual tokens. A higher reconstruction loss indicates that information encoded in the pre-projection representation is harder to recover from the projected representation.

We use this reconstruction loss as a representation-level measure of information recoverability. However, it does not directly measure semantic degradation: some unrecoverable information may be redundant or irrelevant for caption generation. We treat reconstruction loss as an embedding-level signal whose relationship to generated captions must be evaluated separately.

### 3.4 Caption Semantic Evaluation

Representation-level distortion does not directly imply semantic degradation in generated language. We therefore analyze whether embedding-level changes are reflected in captions through two complementary probes.

First, we relate reconstruction loss to caption semantics. For each image, we generate a caption from the original projected representation and score it with BERTScore (Zhang* et al., 2020) and CLIPScore (Hessel et al., 2021), capturing text-level semantic similarity and image-text alignment, respectively. This examines whether images with higher recoverability loss tend to produce semantically weaker or less visually aligned captions.

Second, we conduct controlled perturbation experiments. We modify visual representations before caption generation and compare the captions generated from original and perturbed representations. This directly tests whether representation changes induce semantic changes in the output.

## 4 Experimental Setup

### 4.1 Models and Dataset

We evaluate LLaVA, IDEFICS2, Qwen2.5-VL, and Qwen3.5, the four VLMs discussed in Section 2.1. For each model, we extract pre-projection representations from the vision encoder and post-projection representations from the visual embedding sequence consumed by the language model.

We use the web-sourced subset of the MTF 2025 Dishcovery Dataset, which contains approximately 139K food-domain image–text pairs with single reference captions and was curated using the Precision at Scale method (Rodríguez-de Vera et al., 2025). We treat it as a probing dataset rather than a standard captioning benchmark; therefore, our results should be interpreted as food-domain representation–caption analysis rather than general-purpose captioning evaluation.

### 4.2 Metrics

We use metrics at three levels. For representation geometry, we report the $k$-NN overlap ratio between pre- and post-projection neighborhoods, along with CLIP-based neighborhood overlap as an external semantic reference. For representation recoverability, we use the reconstruction loss defined in Section 3.3, where higher loss indicates that pre-projection information is harder to recover from the projected representation.

For caption-level behavior, we use BERTScore (Zhang* et al., 2020) and CLIPScore (Hessel et al., 2021). BERTScore measures semantic similarity between generated captions and reference captions, while CLIPScore measures image-text alignment. Since each image has only one reference caption, we treat these metrics as relative semantic signals rather than definitive caption-quality scores.

For perturbation experiments, we generate two captions for each image: $y_i$ from the original visual representation and $\tilde{y}_i$ from the perturbed representation. We use BERTScore between $y_i$ and $\tilde{y}_i$ to measure caption stability, where a lower score indicates a larger semantic change. To measure changes in image-text alignment, we compute the difference in CLIPScore before and after perturbation:

$$\Delta\text{CLIPScore}_i = \text{CLIPScore}(x_i, y_i) - \text{CLIPScore}(x_i, \tilde{y}_i).$$

Positive values indicate a decrease in CLIP-based image-text alignment after perturbation, while values near zero indicate little change.

### 4.3 Baselines and Controls

For neighborhood analysis, the main baseline is the pre-projection representation. Comparing pre- and post-projection neighborhoods measures how much the connector changes the original vision-encoder geometry. CLIP-based neighborhoods are used only as an external semantic reference, not as ground-truth semantic labels.

For reconstruction analysis, reconstruction loss is compared against caption-level metrics rather than interpreted directly as semantic degradation. This controls for the possibility that information unrecoverable from the projected representation may be redundant or irrelevant for caption generation.

For perturbation experiments, captions generated from the original post-projection representations serve as the control condition. Captions generated from perturbed representations are then compared against this control to measure caption stability and changes in image-text alignment.

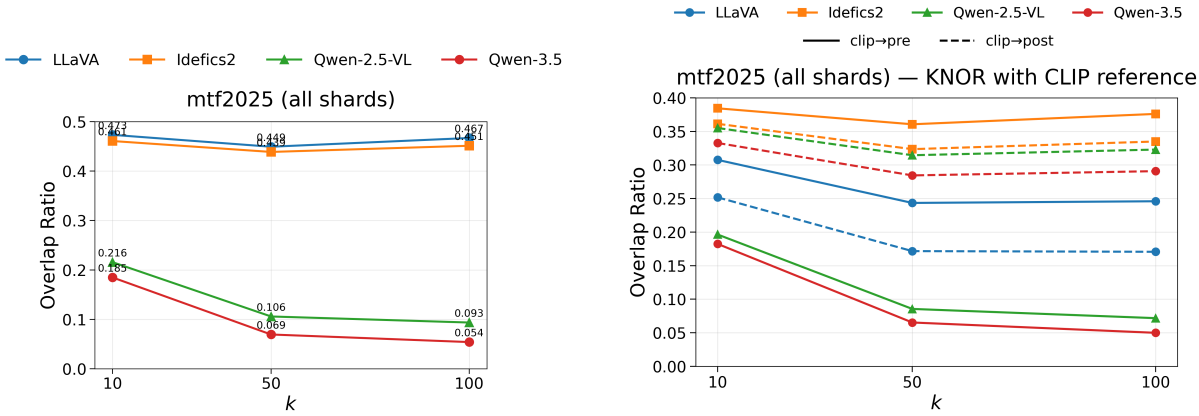
## 5 Results

We organize the results around the three analyses described above: local neighborhood changes, reconstruction-based recoverability, and controlled perturbations. The main question is whether these representation-level signals align with caption-level semantic behavior.

### 5.1 Local Geometry Changes After Projection

We first examine whether connector projection preserves local neighborhood structure. Figure 1a shows the pre/post $k$-NN overlap across models. LLaVA and IDEFICS2 retain substantially higher neighborhood overlap after projection, whereas Qwen-2.5-VL and Qwen-3.5 exhibit much lower overlap. This indicates stronger local geometry reordering in the Qwen-family models.

However, low pre/post overlap does not necessarily imply semantic degradation. As shown in Figure 1b, the relationship to CLIP-based neighborhoods differs across models. For LLaVA and IDEFICS2, the pre-projection space is more aligned with CLIP-based neighborhoods than the post-projection space. In contrast, for Qwen-2.5-VL and Qwen-3.5, the post-projection space is substantially more aligned with CLIP-based neighborhoods than the pre-projection space. This is consistent with the idea that Qwen-family models use projection as a stronger representation transformation, rather than simply preserving a frozen visual




(a) Pre/post overlap

(b) CLIP-reference overlap

Figure 1: Local neighborhood analysis. Pre/post overlap measures how much connector projection preserves the original vision-encoder geometry. CLIP-reference overlap compares pre- and post-projection neighborhoods with CLIP-based semantic neighborhoods.

feature space. Hence, their large pre/post changes should not be interpreted as semantic degradation by default.

Overall, connector projection substantially alters local representation geometry, but the meaning of this change depends on the model architecture. Large pre/post distortion alone is therefore not sufficient to conclude that the projected representation is semantically worse.

### 5.2 Reconstruction Loss and Caption-Level Semantics

We next examine whether reconstruction quality is reflected in caption-level behavior. Table 1 reports Pearson correlations, with Spearman correlations shown in parentheses, between reconstruction metrics and caption-level metrics. Let $y_i^{\text{orig}}$ and $y_i^{\text{recon}}$ be captions generated from the original and reconstructed representations, respectively. We define CLIPScore Drop as

$$\Delta C_i = \text{CLIPScore}(x_i, y_i^{\text{orig}}) - \text{CLIPScore}(x_i, y_i^{\text{recon}}),$$

and caption stability as

$$B_i = \text{BERTScore}(y_i^{\text{recon}}, y_i^{\text{orig}}).$$

Higher $\Delta C_i$ indicates a larger decrease in CLIP-based image-text alignment, while higher $B_i$ indicates greater similarity between the reconstructed and original captions.

Table 1 shows that the relationship is model-dependent. LLaVA shows near-zero correlations across all metrics, suggesting that reconstruction quality explains little of the caption variation in this model. IDEFICS2 shows a clearer relationship with caption stability: higher reconstruction cosine is associated with higher $B_i$, while higher MSE is associated with lower $B_i$. Qwen-2.5-VL again shows weak correlations. In contrast, Qwen-3.5 shows a stronger relationship for reconstruction cosine: higher cosine is associated with lower $\Delta C_i$ and higher $B_i$. However, its MSE-based correlations remain weak.

Figure 2 visualizes the relationship between post-space reconstruction cosine and $\Delta C_i$. Qwen-3.5 shows the clearest negative trend, while the other models exhibit weaker or more diffuse patterns. However, the spread of $\Delta C_i$ among samples with similar cosine values suggests that reconstruction similarity alone does not fully explain caption-level changes.

Table 2 gives representative examples. In some cases, reconstruction changes fine-grained visual details while preserving the broad scene; in others, it mainly produces paraphrases or minor omissions. These examples illustrate why caption-level effects cannot be inferred from reconstruction cosine alone.

Overall, reconstruction-based recoverability sometimes reflects caption-level stability, but the relationship varies across models and metrics. This suggests that reconstruction loss captures an important representation-level change, but only part of that change is relevant to caption generation. Therefore, we treat reconstruction-based distortion as a useful diagnostic signal rather than a direct measure of semantic degradation.

| Model | cos→CLIP↓ | MSE→CLIP↓ | cos→BERT | MSE→BERT |
|---|---|---|---|---|
| LLaVA | -0.001 (0.037) | 0.013 (-0.039) | 0.042 (0.014) | -0.041 (-0.007) |
| IDEFICS2 | -0.069 (-0.068) | 0.065 (0.059) | 0.249 (0.278) | -0.272 (-0.290) |
| Qwen-2.5-VL | -0.056 (-0.043) | -0.017 (-0.011) | 0.124 (0.118) | 0.006 (0.028) |
| Qwen-3.5 | -0.461 (-0.356) | -0.081 (-0.026) | 0.375 (0.437) | 0.049 (0.043) |

Table 1: Correlations between reconstruction quality and caption-level behavior. Values are Pearson $r$ with Spearman $\rho$ in parentheses. CLIP↓ is original CLIPScore minus reconstructed-caption CLIPScore, so higher values indicate larger CLIPScore drops. BERT measures similarity between reconstructed and original captions.

| Model / case | Original caption | Reconstructed caption | Metrics |
|---|---|---|---|
| Qwen-3.5 / high drop | A bowl of golden-brown ramen broth with chashu pork, egg yolk, green onions, and Japanese text indicating a "Saga Ramen" set. | A bowl of clear Asian soup with dumplings or meatballs, green onions, and an egg yolk. | $\Delta C = 5.60$ cos$= 0.833$ $B = 0.732$ |
| Qwen-3.5 / low drop | A metal bowl of bibimbap on a rustic wooden table with side dishes such as kimchi and pickled radish. | A traditional Korean bibimbap in a black stone bowl with seasoned vegetables and side dishes. | $\Delta C = 0.05$ cos$= 0.913$ $B = 0.838$ |
| LLaVA / small drop | A plate of food with rice, meat, vegetables, a bowl of sauce, and a piece of bread on the side. | A plate of food with rice, meat, and sauce. | $\Delta C = 0.84$ cos$= 0.984$ $B = 0.959$ |
| IDEFICS2 / low cosine | A plate containing food items and a plant. | A plate containing food. | $\Delta C = 0.23$ cos$= 0.660$ $B = 0.981$ |

Table 2: Qualitative caption comparisons after reconstruction. Red and Green text highlights details that are changed, removed, or newly introduced after reconstruction. $\Delta C$ denotes CLIPScore Drop and $B$ denotes BERTScore between captions generated from original and reconstructed representations.

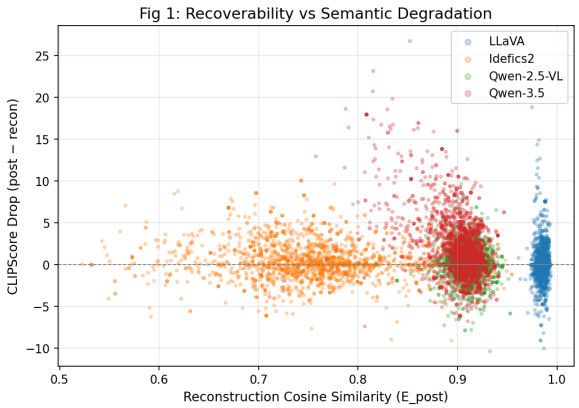



Figure 2: Post-space reconstruction cosine similarity versus CLIPScore Drop. All observed cosine values are above 0.5. Qwen-3.5 shows a clearer negative trend, while other models show weaker or more diffuse relationships, indicating that reconstruction similarity does not consistently explain caption-level changes across models.

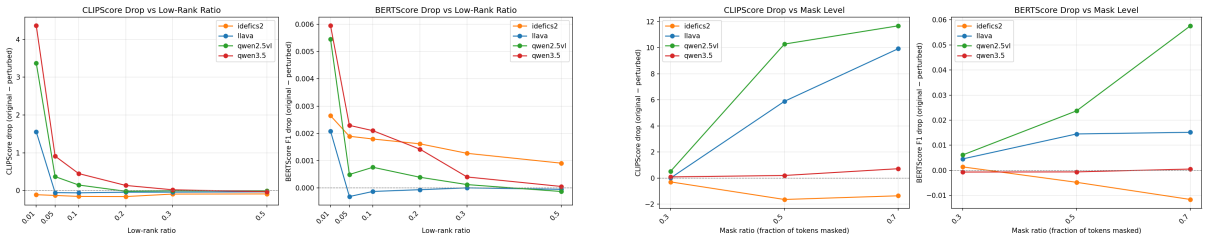
### 5.3 Controlled Perturbation Analysis

Finally, we directly perturb visual representations before caption generation. We use low-rank truncation to keep only dominant representation directions, and token masking to remove visual tokens. We report CLIPScore Drop and BERTScore Drop, defined as original score minus perturbed score, where larger positive values indicate larger metric decreases.

Figure 3 shows that low-rank perturbation mainly affects models when the retained-rank ratio is very small. This pattern is consistent with the spectral diagnostics in Table 3. The normalized effective rank $r_{\text{eff}}^{post}/R_{\max}$ indicates the fraction of post-projection rank needed to represent the effective dimensionality of the projected space. Qwen-2.5-VL and Qwen-3.5 have normalized effective ranks of 0.213 and 0.125, so the smallest retained ratios fall far below their effective dimensionality and produce large drops. In contrast, LLaVA has a highly concentrated post-projection spectrum, with normalized effective rank 0.006 and 96.9% top-10 energy, which is consistent with its smaller low-rank sensitivity.

Token masking shows a different sensitivity pattern. Qwen-2.5-VL and LLaVA show larger drops as the mask ratio increases, while IDEFICS2 and Qwen-3.5 are less affected. This suggests that spectral perturbations and token-level perturbations remove different types of information. However, smaller drops should not be interpreted as stronger semantic robustness by default, since they may also reflect lower baseline caption scores or reduced sensitivity of the chosen metrics.

Overall, perturbation results show that representation changes can affect caption-level behavior,




(a) Low-rank perturbation

(b) Token masking

Figure 3: Caption-level changes under controlled perturbations. CLIPScore Drop and BERTScore Drop are defined as original score minus perturbed score, so larger positive values indicate a larger decrease under the corresponding metric. Low-rank perturbations mainly affect Qwen-family models at very small retained-rank ratios, while token masking produces stronger model-dependent effects, especially for Qwen-2.5-VL and LLaVA.

| Model | $r_{\text{eff}}^{post}$ | Norm. $r_{\text{eff}}$ | Top-10 energy |
|---|---|---|---|
| LLaVA | 3.41 | 0.006 | 0.969 |
| IDEFICS2 | 26.76 | 0.418 | 0.644 |
| Qwen-2.5-VL | 30.70 | 0.213 | 0.661 |
| Qwen-3.5 | 12.51 | 0.125 | 0.812 |

Table 3: Post-projection spectral diagnostics. Normalized effective rank is computed as $r_{\text{eff}}^{post}/R_{\max}$ and is directly comparable to the retained-rank ratios used in the low-rank perturbation experiment. Top-10 energy reports the fraction of spectral energy captured by the top 10 directions.

but the effect is model- and perturbation-dependent. Effective rank helps explain low-rank sensitivity, but it is not a complete explanation. For example, IDEFICS2 has the largest normalized effective rank but does not show the largest drop, possibly because its baseline caption-level scores are already lower, leaving less room for additional decreases under perturbation.

## 6 Discussion and Limitations

Our results suggest that connector-induced representation distortion should not be interpreted as semantic degradation by default. The kNN analysis shows that projection can substantially reorder local neighborhoods, especially for Qwen-family models. However, the CLIP-based neighborhood comparison indicates that a low pre/post overlap may also reflect a reorganization of the representation space rather than a simple loss of semantic structure. This is important because the pre-projection representation is only a reference space, not an absolute ground truth for visual semantics.

The reconstruction results further support this distinction. Reconstruction quality is sometimes associated with caption-level behavior, but the relationship is not consistent across models or metrics. For example, IDEFICS2 and Qwen-3.5 show some meaningful correlation between reconstruction cosine and caption stability, while LLaVA and Qwen-2.5-VL show weaker patterns. This suggests that reconstruction-based recoverability is a useful diagnostic signal, but not a universal explanation of downstream semantic behavior. In particular, cosine similarity and MSE can lead to different conclusions, implying that different distortion metrics capture different aspects of representation change.

The perturbation experiments provide a more direct test of caption sensitivity. Low-rank perturbation shows that caption-level behavior can change sharply when the retained rank falls below the effective dimensionality of the projected representation. Token masking shows a different pattern, suggesting that spectral perturbations and token-level perturbations affect different types of information. However, smaller drops should not be interpreted as stronger semantic robustness by default. They may also reflect lower baseline caption scores, metric insensitivity, or the possibility that the perturbation does not target information used by the language model.

There are several limitations. First, the MTF 2025 web-sourced subset is a food-domain probing dataset, not a standard captioning benchmark. Since most images are food-related and each image has only one reference caption, our results should be interpreted as domain-specific representation–caption analysis. Second, CLIP-based neighborhoods, CLIPScore, and BERTScore are diagnostic signals, not ground-truth semantic judgments. Third, reconstruction loss depends on reconstructor capacity and may reflect both connector-induced information loss and reconstruction limitations. Finally, our perturbations are not exhaustive; other perturbations may reveal different sensitivities.

## 7 Conclusion

This work analyzed whether connector-induced representation distortion is reflected in caption-level semantic behavior in VLMs. We compared pre- and post-projection representations using local neighborhood overlap, reconstruction-based recoverability, and controlled perturbation experiments.

A key finding is that large representation distortion does not have a single interpretation across models. Qwen-family models show strong pre/post geometry changes, but their post-projection spaces can align more closely with CLIP-based neighborhoods than their pre-projection spaces. In addition, Qwen3.5 shows the strongest association between reconstruction cosine and caption-level behavior, while its MSE-based correlations remain weak. This suggests that directional similarity in the projected representation may matter more for caption stability than raw Euclidean reconstruction error in some models.

Overall, our results show that connector projection can substantially alter local representation geometry, but such changes do not always imply semantic degradation. Reconstruction-based metrics sometimes correlate with caption-level stability, and perturbation experiments show that captions can be sensitive to representation changes. However, these effects are model-dependent and metric-dependent. Future work could study controlled connector-training settings using differentiable surrogate objectives or reinforcement-learning-style optimization to better test whether geometry preservation and caption-level behavior can be jointly maintained.

## Contribution Statement

This project was completed individually. I was responsible for all parts of the project, including problem formulation, literature review, model selection, representation extraction, reconstruction experiments, kNN analysis, perturbation experiments, metric evaluation, result visualization, and writing the final report. I used AI tools only for language polishing, organization suggestions, and debugging assistance; all experiments, analysis decisions, reported results, and citations were checked by me.

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